

Customer Personality Analysis

Tasks 10–13 Technical & Business Analysis Report

Prepared By: Muhammad Haider Iqbal

Project: Machine Learning Internship

Scope: Tasks 10 – 13

TASK 10 – BUSINESS UNDERSTANDING

Objective

The primary objective of this analysis is to thoroughly understand customer behavior, purchasing patterns, and marketing campaign performance. The dataset provides key demographic information, purchasing history across various channels, website engagement metrics, and campaign responses. These insights empower businesses to make data-driven decisions, enhance customer engagement, optimize marketing budgets, and drive sustainable revenue growth.

Business Indicators

Business Indicator	Explanation & Business Impact
High Income Customers	Customers with higher annual income possess greater purchasing power and display a higher propensity to buy premium offerings. They are critical drivers for top-line revenue growth and margin expansion.
Frequent Buyers	Customers who make frequent purchases via physical stores, web, or catalogs represent loyal repeat clients. They are ideal targets for tiered loyalty programs, VIP perks, and tailored upselling.
Campaign Responders	Customers who accepted previous promotional campaigns exhibit high marketing receptivity. Targeting this segment maximizes ROI and conversion rates on upcoming promotional drives.
Customers with Complaints	Tracking active complaints helps identify operational bottlenecks and service quality deficits. Promptly addressing these issues protects customer lifetime value and reduces churn.
Website Visitors	Frequent digital visits signal high engagement. Monitoring online activity allows evaluation of digital UX, web content resonance, and conversion funnel optimization.
High Spending Customers	Clients generating high spend across multiple product categories contribute heavily to core revenue. They require personalized recommendations and dedicated relationship management.
Recently Active Customers	Customers with low Recency scores have purchased recently and are highly warm leads for new product launches, seasonal promotions, and cross-channel marketing.

Business Summary

This dataset forms the foundation for data-driven customer relationship management (CRM). By synthesizing demographic profiles, historical spending patterns, and promotional responses, businesses can implement granular customer segmentation, personalize outreach, improve customer retention, and maximize return on marketing investments.

TASK 11 – INITIAL DATA SUMMARY REPORT

Dataset Overview

The Customer Personality Analysis dataset compiles customer demographics, buying behaviors, marketing campaign outcomes, complaint logs, and web navigation statistics. It is specifically designed for unsupervised learning (clustering) and supervised predictive modeling.

Dataset Information

Property	Value
Dataset Name	Customer Personality Analysis
File Format	CSV
Number of Records	2,240
Number of Variables	29
Dataset Size	215.03 KB

Duplicate Records Check

Check Type	Result	Status
Duplicate Rows	0	Passed
Duplicate Customer IDs	0	Passed
Negative Income Check	0	Passed

Missing Values Analysis

Variable Name	Missing Count	Missing Percentage	Impact & Status
Income	24	1.07%	Minor missingness; suitable for median imputation.
All Other Variables (28)	0	0.00%	Fully populated.

Numerical Variables (26)

ID

Year_Birth

Income

Kidhome

Teenhome

Recency

MntWines

MntFruits

MntMeatProducts

MntFishProducts

MntSweetProducts

MntGoldProds

NumDealsPurchases

NumWebPurchases

NumCatalogPurchases

NumStorePurchases

NumWebVisitsMonth

AcceptedCmp1

AcceptedCmp2

AcceptedCmp3

AcceptedCmp4

AcceptedCmp5

Complain

Z_CostContact

Z_Revenue

Response

Categorical Variables (3)

Education

Marital_Status

Dt_Customer

Major Data Quality Observations

- **Income Missingness:** 24 records (1.07%) have missing income values that need handling during preprocessing.
- **Age Anomaly:** Calculated maximum customer age reaches ~133 years (based on Year_Birth), representing an obvious extreme outlier.
- **Income Outlier:** Maximum income reaches 666,666, which deviates significantly from the median and requires cap/ treatment.
- **Unconventional Categoricals:** Marital_Status contains non-standard labels like "Alone", "Absurd", and "YOLO".
- **Constant Features:** Z_CostContact and Z_Revenue contain identical values across all 2,240 rows, adding zero variance.

Initial Conclusion

The dataset is structural, clean, and highly usable. Standard preprocessing tasks (handling 1.07% missing values, filtering outliers, consolidating rare categories, and dropping constant columns) will render it fully optimized for modeling.

TASK 12 – DATA DICTIONARY

Variable	Description	Data Type	Business Meaning
ID	Unique customer identification number	Integer	Identifies each customer uniquely
Year_Birth	Customer's year of birth	Integer	Calculates customer age
Education	Level of education attained	String	Reflects educational background
Marital_Status	Marital / household status	String	Indicates family/marital structure
Income	Annual household income	Float	Measures purchasing power
Kidhome	Number of small children in household	Integer	Indicates presence of young kids
Teenhome	Number of teenagers in household	Integer	Indicates presence of teenagers
Dt_Customer	Date customer enrolled with company	Date (String)	Tracks relationship tenure
Recency	Days since last customer purchase	Integer	Measures recent transaction activity
MntWines	Amount spent on wine products	Integer	Wine product revenue contribution
MntFruits	Amount spent on fruit products	Integer	Fruit product revenue contribution
MntMeatProducts	Amount spent on meat products	Integer	Meat product revenue contribution
MntFishProducts	Amount spent on fish products	Integer	Fish product revenue contribution
MntSweetProducts	Amount spent on sweet products	Integer	Sweets product revenue contribution
MntGoldProds	Amount spent on gold/premium items	Integer	Premium product revenue contribution
NumDealsPurchases	Purchases made with a discount	Integer	Evaluates sensitivity to discounts
NumWebPurchases	Purchases made through company website	Integer	Digital channel purchasing volume
NumCatalogPurchases	Purchases made using catalog	Integer	Direct catalog response rate
NumStorePurchases	Purchases made directly in stores	Integer	Physical store transaction volume
NumWebVisitsMonth	Number of website visits last month	Integer	Online engagement frequency
AcceptedCmp1–5	Accepted campaign 1 to 5 (binary)	Integer	Historical response to 5 campaigns
Complain	1 if customer complained in last 2 yrs	Integer	Customer dissatisfaction indicator
Z_CostContact	Cost to contact customer (constant)	Integer	Fixed operational contact cost
Z_Revenue	Revenue benchmark (constant)	Integer	Fixed benchmark revenue unit

Variable	Description	Data Type	Business Meaning
Response	1 if accepted final campaign offer	Integer	Target variable for response modeling

TASK 13 – DATA QUALITY REPORT

Objective

The objective of this Data Quality Report is to systematically identify data quality issues prior to feature engineering and statistical modeling, and to prescribe actionable remediation steps to ensure data integrity.

Data Quality Assessment & Recommendations

Issue Category	Variable	Description	Recommended Action
Missing Values	Income	24 missing values detected (1.07% of total records).	Impute missing values using median income group or iterative imputer.
Duplicate Rows	Entire Dataset	No duplicate rows identified across all 2,240 records.	No action required.
Duplicate IDs	ID	No duplicate customer IDs detected.	No action required.
Negative Values	Income	No negative values observed in annual income.	No action required.
Age Outlier	Year_Birth	Calculated maximum age is ~133 years (birth year 1893).	Filter out records where birth year < 1940 (age > 80–90).
Income Outlier	Income	Extreme income peak at 666,666.	Cap at 99th percentile or remove rows exceeding \$100,000 threshold.
Uncommon Categories	Marital_Status	Unusual categories: "Alone", "Absurd", and "YOLO".	Group "Alone" under "Single", and remove/recode "Absurd" & "YOLO".
Date Format	Dt_Customer	Enrollment dates stored as text string objects.	Convert to datetime and derive "Customer_Tenure_Days".
Constant Features	Z_CostContact, Z_Revenue	Zero variance; identical value in every record.	Drop both variables during feature selection stage.

Data Quality Executive Summary

Quality Metric / Check	Result / Finding
Total Records Processed	2,240
Total Variables Inspected	29
Columns Containing Missing Values	1 (Income)
Total Missing Value Count	24 (1.07%)

Quality Metric / Check	Result / Finding
Duplicate Rows Found	0
Duplicate Customer IDs Found	0
Negative Income Values Detected	0
Potential Age Outlier Flagged	Yes (>100 years old)
Potential Income Outlier Flagged	Yes (\$666,666)
Uncommon Categorical Labels Flagged	Yes (Alone, Absurd, YOLO)
Zero Variance / Constant Variables	2 (Z_CostContact, Z_Revenue)

Final Preprocessing & Modeling Readiness Conclusion

The Customer Personality Analysis dataset demonstrates high overall structural integrity with zero duplicate records and minimal missing data. The execution of the recommended remediation steps—imputing missing income, trimming extreme age/income outliers, recoding rare marital categories, converting string dates into numerical customer tenure, and removing zero-variance constant features—will yield a clean, robust, and highly reliable dataset for clustering, segmentation, and machine learning models.